

Terraform Smoke Test Plan

AI Cloud Lab | From variable setup to a working private Open WebUI tunnel

Expected result	Terraform should run if AWS access, quota, a default VPC, and the selected EC2 instance type are available.
Typical runtime	Allow roughly 10-20 minutes of instance runtime for bootstrap and model download.
Cost reminder	The default c7i.4xlarge is expensive. Stop it immediately after testing and destroy it when finished.

1. Set the test variables

Set the AWS profile, region, and temporary Open WebUI password before running Terraform. The password is required and has no default.

```
# Use the profile and region you verified
$env:TF_VAR_aws_profile = "ai-cloud-lab"
$env:TF_VAR_aws_region = "us-east-1"
$env:TF_VAR_open_webui_admin_password = "YourTemporaryStrongPasswordHere"
```

Alternatively, pass non-secret variables directly to Terraform. Keep the password in the environment when possible.

```
terraform plan `
  -var="aws_profile=ai-cloud-lab" `
  -var="aws_region=us-east-1"
```

2. Validate and preview

If the plan fails, check credentials, EC2 quota, the default VPC, and c7i.4xlarge availability in the selected availability zone.

```
terraform init
terraform fmt -check -recursive
terraform validate
terraform plan
```

3. Deploy

Apply returns when EC2 is running, not necessarily when the model and Open WebUI are ready. Bootstrap continues in the background.

```
terraform apply
```

4. Check readiness through SSM

```
terraform output -raw instance_id

aws ssm start-session --target <instance-id> --region us-east-1 --profile ai-cloud-lab
```

Inside the SSM shell:

```
if test -f /var/lib/ai-lab/ready; then echo READY; elif test -f /var/lib/ai-lab/failed; then echo FAILED;
tail -n 100 /var/log/ai-lab-bootstrap.log
ai-lab-status
```

5. Open the private browser tunnel

Run the printed command in a second terminal, then open <http://localhost:8080>.

```
terraform output -raw ssm_open_webui_port_forward_command
```

6. Dismantle the test system

Stop the instance if you may test again later. Destroy all Terraform-managed infrastructure when finished. Review the destroy plan before confirming.

```
# Temporary pause
aws ec2 stop-instances --instance-ids <instance-id> `
  --region us-east-1 --profile ai-cloud-lab

# Permanent teardown
terraform plan -destroy
terraform destroy
```

After teardown, verify the instance and security group are gone. Remove local terraform.tfvars if it contains a real password, but keep terraform.tfvars.example.

Likely failure points

- Missing AWS permissions or expired SSO login.
- No default VPC in the selected region.
- Insufficient c7i.4xlarge quota or AZ availability.
- Missing Session Manager plugin.
- Bootstrap still downloading packages, Ollama, the model, or Open WebUI.